Data Canvas: A Provenance-Guided Harness for Agentic Data Engineering

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ABSTRACT

As agentic LLM-based technologies mature, they hold tremendous promise for designing and executing complex *data engineering* tasks—compiling and integrating specialized datasets on behalf of users—spanning high-level planning and query optimization down to runtime extraction and semantic reasoning. Recent frontier systems show strong potential but remain prone to hallucinations, omissions, and underspecified decisions that are hard to inspect or correct. We argue that agentic data engineering needs a provenance-guided harness: a control layer around planning and execution that makes outputs attributable, inspectable, and steerable through feedback. We present Data Canvas, which wraps LLM-driven execution with structured semantic operators, explicit intermediate tables, and a provenance graph over operator events and results. It supports sparse human or automated feedback by tracing errors to their responsible reasoning steps, propagating corrections to related outputs, and replaying only the affected portions of the workflow. On standard deep research benchmarks, Data Canvas improves answer quality over existing systems, and provenance-guided feedback yields substantial gains at a small fraction of the cost of prompt re-engineering and re-planning.

1 INTRODUCTION

Agentic LLM-based technologies promise a new wave of *agentic data engineering* tools (e.g., [1, 19]), analogous to agentic coding tools but oriented around data synthesis: extracting structured data from unstructured sources, mapping heterogeneous data to a schema, and cleaning and linking data. Tasks traditionally handled by ETL tools or crowdworkers (e.g., [20]) can now be tackled autonomously at runtime.

Example 1. *Suppose we want to help prospective CS graduate students identify PhD advisors. A first step is to build a dataset of all CS faculty who can advise students across R1 universities in North America, with their research areas. A deep research system may work, but may also: (1) give incomplete answers, (2) hallucinate, and (3) resolve ambiguity (e.g., whether to include emeritus faculty) in unspecified ways, without consulting the user.*

Compared to deep research systems (e.g., [15]), agentic data engineering approaches develop a more regularized, plan-based strategy for assembling answers over large source collections: high-level planning via large reasoning models and semantic query optimization [7, 18], decomposition into table-creation operators for "wide search" [4, 11, 22], then execution via lower-level primitives for extraction [1, 19], map-reduce [4], and semantic reasoning over results [17]. Because expectations for soundness and completeness are high, there is an unmet need for users to **understand, assess, and repair model mistakes**. We argue this requires a harness: a control layer around planning and execution that explains and attributes answers, accepts feedback on output quality, and adapts reasoning traces dynamically. Inspired by the database provenance and view-maintenance literature, we combine user feedback with how answers were derived to efficiently repair errors.

- Data Canvas, a provenance-guided harness producing regular, structured LLM-driven plans with semantic operators for data engineering tasks.
- Strategies for tracking provenance as a graph and querying result derivations across plan steps back to their sources.
- A method to incorporate sparse feedback by propagating "blame" across the provenance graph to specific reasoning steps.
- On standard benchmarks, our approach improves answer quality at a fraction of the cost of prompt re-engineering and full re-execution.

2 RELATED WORK

Large reasoning models trained with RLHF [16] have led to "deep research" systems [7, 23] for complex information-gathering, but their search strategies limit exploration, yielding incomplete answer sets. In contrast, WideSearch [22] targets answer-assembly with regular plans but greater *breadth*: given a query and schema, an agent exhaustively populates a results table from live web sources. DeepWideSearch [12] further combines breadth with multi-hop depth. Recent refinements include TaS [11] (table completion), WideSeek [9] (RL-trained multi-agent policies), and A-MapReduce [4] (parallel allocation with experiential memory). SODIUM [8] targets *depth within a domain.*

Data Canvas operates in the WideSearch setting—open-web, multi-source, schema-driven extraction—and creates plans from *semantic operators* [1, 13, 17, 18], which we connect to *provenance.* In general, an agentic plan resembles a workflow pipeline, captured by the PROV [2] standard. The database literature developed a query-semantics-preserving treatment of provenance over relational operators: "why provenance" [3] and provenance semirings [6] capture both alternate derivations and derivation from combined inputs, supporting incremental updates [5]. We adapt this to the agentic setting, tracing incorrect outputs back to reasoning steps and sources and performing *incremental view maintenance* [14] to propagate corrections—more targeted than global RL and more cost-effective than prompt-update recomputation, as in DSPy [10]. Finally, Data Canvas's provenance and feedback mechanisms can support more collaborative human–agent pipelines for maintaining tabular datasets [20].

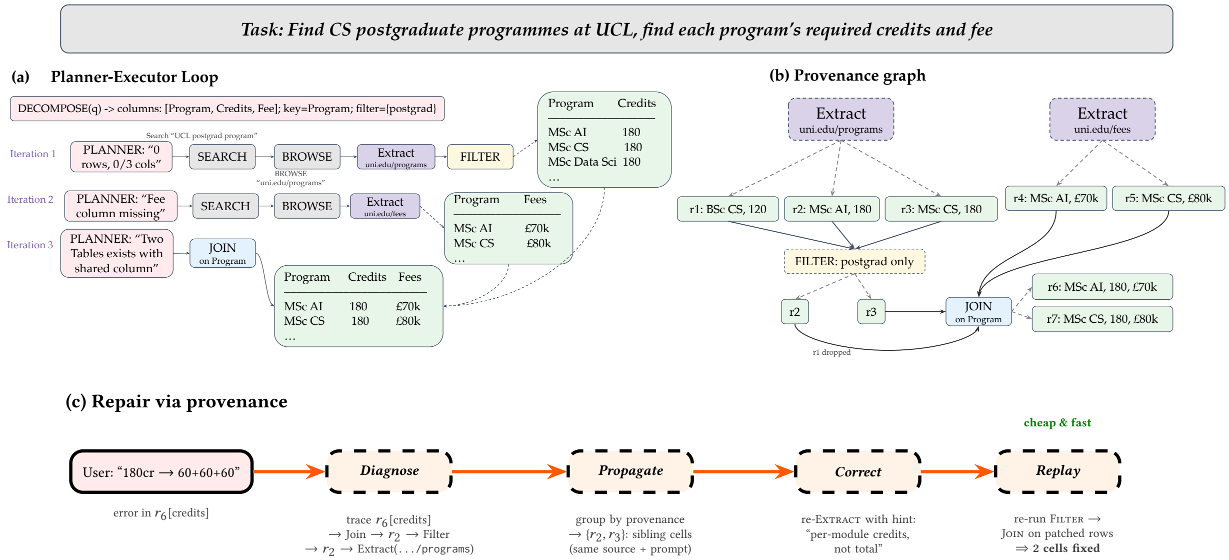

Figure 1: Provenance graph and repair. (a) The Planner-Executor loop builds a result table through operators and intermediate tables. (b) The provenance graph records how rows are derived from sources and operators. (c) Provenance enables targeted repair: a corrected cell is traced, propagated, and replayed to fix related errors—whereas without provenance, the only option is to append the correction to the query and re-run the entire pipeline.

3 DATA CANVAS

Data Canvas adopts a ReAct-based approach [23]: it iteratively decomposes a prompt into actions via a receding-horizon plan–execute loop, where each action is a semantic operator from a small library (Table 1). Each operator reads the tabular results of prior operators and produces its own. These intermediate relations are named artifacts in the plan state, materialized outside the LLM context as persisted snapshots. Unlike prior semantic operators [17–19], **each of our operators records provenance**: any output can be *algorithmically* tied to its operations and inputs.

Example 2. *Figure 1(a) shows how Data Canvas decomposes a task ("Find CS postgraduate programmes at UCL; find each program's required credits and fee") into semantic operators. Iteration 1 searches for pages, browses to extract text, and extracts program credits; Iteration 2 runs a similar pipeline for fees; Iteration 3 joins the results. Provenance recorded during execution yields the graph in Figure 1(b).*

3.1 Iterative Planning and Execution in Data Canvas

The system first determines a target schema (columns, keys, filters) and reformulates the prompt accordingly. At each iteration the planner reasons over the *relational intermediate state* (artifact schemas, row counts, entity samples, and per-column coverage), selects one operator and its arguments, and observes the result. This enables targeted decisions: eg. missing columns trigger Map sub-agents for per-entity enrichment; shared keys trigger a Join; diminishing returns trigger termination. The executor then dispatches the operator, records provenance, and appends to the step history. When the planner invokes Map, it partitions the working set and recursively runs a full plan-execute loop on each part in parallel, then Reduces the sub-results and merges their provenance subgraphs. The loop ends when the planner stops with no coverage gaps, or upon stagnation (no new rows or filled cells).

Table 1: Semantic operators in Data Canvas.

Search(q)	issues web search q; returns URLs with snippets.
Browse(u)	fetches URL u and returns its page text.
Extract(D, S)	extracts structured rows from documents D for schema S.
Filter(σ, R)	retains rows of R matching predicate σ (rule-based or LLM-semantic).
Project(R, Π)	returns only the columns of R in Π.
Merge(R, K)	deduplicates R on keys K, using an LLM to resolve conflicts.
Join(R_1, R_2, K)	joins on shared keys K, with optional LLM fuzzy matching.
Union, Sort	standard relational operators.
Map(f, R)	launches parallel sub-executors (per row/column), each running its own planning loop to enrich R.

3.2 Provenance in Data Canvas

We represent provenance as a *directed acyclic graph* $\mathcal{G}$ of *data nodes* interleaved with *event nodes* (Figure 1b). A data node holds a *row ID* referencing the on-disk CSV snapshot of an operator's output; an event node is a single operator invocation, with arcs from the rows it consumed to those it emitted. Each event records a *forward row mapping* from every input row to the outputs derived from it (dropped rows map to $\perp$), plus its operator type and source metadata. Tuple-to-tuple operators (Filter) map inputs one-to-one or to $\perp$; combining operators (Merge, Join) map many contributing inputs to one output, capturing many-to-one derivations. The graph is built incrementally and browsed pages are cached by URL for reproducible repair.

The graph supports two query patterns over these mappings. *Trace queries* reconstruct a cell's derivation by walking backward and inverting the forward mapping at each step (e.g., r_6[credits] traces through JOIN and FILTER to EXTRACT_1 at source `uni.edu/programs`). *Impact queries* use an event's forward mapping to enumerate the *sibling rows* it produced from the same content, which likely share the same error.

3.3 Strategies for Provenance-Based Propagation of Feedback

Errors in LLM-based execution are *systematic*: the same source processed with the same prompt yields the same mistake for every row with matching provenance. Our repair (Figure 1c) exploits this in three steps. **Diagnose:** we walk backward from the error cell to the leaf BROWSE–EXTRACT; an LLM, given the trace, predicted and gold values, and cached source, identifies the responsible node (e.g., a bad MERGE conflict, an over-aggressive FILTER, or an irrelevant source) and emits a *correction hint.* **Propagate:** the faulty node's forward mapping enumerates its *sibling rows*, which likely share the bug. **Correct and replay:** we patch the node—re-running the operator with the hint on sibling rows, or removing its output rows for source-level exclusion—then re-execute downstream nodes in topological order (running a targeted search–browse–extract for missing entities). Section 4 shows cost reductions up to **35×** vs. re-execution.

4 EXPERIMENTS

We evaluate Data Canvas on two benchmarks against state-of-the-art agentic and deep research systems, addressing: (1) Does structured planning with provenance improve extraction quality? (2) Can lineage-guided sparse feedback improve quality at low cost? (3) How do cost, latency, and quality trade off?

4.1 Experimental setup

Benchmarks. We use the English subset of the WideSearch benchmark [22]: diverse information-seeking queries requiring structured tables extracted from the open web, spanning education, technology, sports, law, business, health, and science. Each query specifies a target schema, with gold tables ranging from 11–533 rows and 3–14 columns. We also evaluate on the Drafty CS Professors dataset [21][1], a curated table of all CS faculty at US, requiring discovery of faculty from departmental pages plus per-attribute extraction—a focused single-institution benchmark for entity-complete extraction.

Metrics. Following the WideSearch protocol, we report *row-level* F1 (whole rows match), *item-level* F1 (per-cell over matched entities), *entity F1* (correct entity set, ignoring attributes), and *cell accuracy.* Matching uses the WideSearch suite with type-specific preprocessing and exact-match or LLM judge.

Baselines. We compare against four systems: **WideSearch** [22], a ReAct-based multi-agent system (best-performing; GPT-5); **Deep-Research** [15], OpenAI's Deep Research, which returns unstructured answers and we applies post-hoc table extraction; **TaS** [11], Table-as-Search, a strong agentic baseline framing seeking as table completion (run with GPT-5); and **TaS (mini)**, a cost-reduced GPT-5-mini variant.

[1]https://drafty.cs.brown.edu/csprofessors

Method	**Row** F1	**Item** F1	**Ent.** F1	**Cell** Acc.	**Lat.** (s)	**Cost** ($)
WideSearch	21.8	43.4	52.6	73.3	830	1.51
DeepResearch	11.7	20.2	23.3	29.2	268	0.83
TaS	25.6	49.6	60.4	73.5	3940	18.27
TaS(mini)	17.1	37.3	49.3	68.8	1835	1.93
Ours	26.6	52.8	63.2	74.0	2276	1.88
Ours+Repair	**28.9**	**55.9**	**66.6**	**74.2**	+235	+0.05

Table 2: Full evaluation on the WideSearch benchmark (F1 reported per granularity; all metrics are percentages). Repair improves quality at negligible cost.

Configuration. Data Canvas uses GPT-5 for planning and GPT-5-mini for extraction, uses Google Search , and Playwright for browsing, replanning each iteration over observed state. For repair, we sample 1% of gold errors as corrections.

4.2 Main results

WideSearch. Table 2 presents the full WideSearch evaluation. Data Canvas (no repair) vs. the strongest baseline WideSearch improves row recall 18.4%⟶24.5% and item recall 36.5%⟶49.0%, a 4.8-point row-F1 gain (26.6 vs. 21.8) and 9.4-point item-F1 gain (52.8 vs. 43.4) at comparable cost ($1.88 vs. $1.51). TaS reaches higher row precision but low recall, at 9.7× the cost and 1.7× the latency; DeepResearch scores only 11.7 row F1, as unstructured systems struggle with schema-conformant extraction.

Drafty. Table 3 reports the Drafty CMU CS Professors task [21]: 7 attributes for all 240 faculty. We omit TaS with GPT-5 here due to prohibitive cost.[2] Data Canvas reaches 75.1% entity F1 and 59.1% item F1, nearly tripling the entity F1 of WideSearch (24.4) and TaS (mini) (24.5). DeepResearch attains 73.8% cell accuracy despite 27.4% entity recall—few rows, but right. Our higher cost ($133.68) reflects per-entity enrichment: attributes like join year and doctoral institution require browsing each professor's profile (~300 subtasks)—a cost the baselines avoid only by achieving much lower entity recall. *Data Canvas discovers nearly 3× more entities, with cost scaling in the number of per-entity page visits.*

4.3 Repair with sparse feedback

We simulate a human-in-the-loop setting: a user corrects a sparse sample of wrong output cells (1% of gold-identified errors), which the repair module then diagnoses, propagates along provenance, corrects, and replays. Diagnosis classifies three common root causes: *extraction ambiguity*, where the extractor picks a wrong interpretation of an underspecified column (e.g., returning total "120" credits when gold wants per-module "60+60"), so one corrected cell yields a hint that propagates to all rows from the same prompt; *filter recovery*, where a too-aggressive filter's dropped rows are re-evaluated against the restored example; and *merge conflict*, re-resolved using the correction as a tie-breaker across the dedup group. Because such errors are systematic, one feedback action can correct tens of results.

[2]TaS with GPT-5 costs $18.27/query for ~20-row tables; the 240-row Drafty task needs far more calls.

Method	Row F1	Item F1	Ent. F1	Cell Acc.	Lat. (s)	Cost ($)
WideSearch	0.2	11.9	24.4	38.5	1823	4.75
DeepResearch	7.0	29.1	37.4	73.8	769	2.77
TaS (mini)	0.7	11.9	24.5	48.2	1588	2.33
Ours	22.7	59.1	74.4	75.1	5914	133.68
Ours+Repair	**25.0**	**64.4**	**81.0**	**75.2**	+5.2	+0.32

Table 3: Results on the Drafty CMU CS Professors task (240 gold rows, 7 columns; F1 per granularity, all percentages). Here repair excludes faculty from computational biology: Data Canvas traces to the shared source and removes the false positives in one shot, improving precision at very low cost.

Results. Table 2 (bottom rows) shows repair improves row F1 26.6→28.9 (+2.3), item F1 52.8→55.9 (+3.1), and entity F1 63.2→66.6 (+3.4), adding only 234.6s and $0.053/query. Without provenance, the only option is to append corrections and re-run the whole pipeline ($1.88, 2276s)—35× costlier and 9.7× slower, with no guarantee the errors are fixed. *Provenance-guided sparse repair yields meaningful gains at* 35× *lower cost than re-execution.*

5 CONCLUSION AND FUTURE WORK

This preliminary work presented Data Canvas, a provenance-guided harness for agentic data engineering, and showed that organizing LLM-agent execution as a regular plan of semantic operators makes its outputs attributable and repairable: sparse user feedback can be traced, propagated, and replayed to correct systematic errors at a small fraction of the cost of full re-execution. Several open questions and challenges remain. One concerns the *reach* of provenance: today we trace errors that are already visible in the output, yet many failures stem from sources that were never retrieved. How far upstream can a provenance model usefully extend—through retrieval, ranking, and query formulation—before lineage becomes too uncertain to act on? A second concerns *feedback*: how should a system generalize from a handful of corrections without overcorrecting, and which feedback is most informative to solicit? A third concerns *generality*. The provenance-based feedback mechanism we aim is not specific to Data Canvas: any agentic data system whose execution can be expressed as regular operators with explicit inputs and outputs—for instance, semantic-operator query engines or structured data-cleaning agents—could adopt the same design by recording which sources, intermediate records, and operator calls produced each output, and could then support a similar diagnose–propagate–replay repair loop.We see Data Canvas as an early step toward agentic data engineering systems that are not only capable but inspectable and steerable, and we leave these questions to future work.

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